

3.2 The Derivative

We now make precise the idea of an instantaneous rate of change.

Definition 3.1. The derivative of f at a is the limit of the difference quotient $(f(a+h) - f(a)) / h$ as h approaches 0, when that limit exists.

Theorem 3.4 (Power Rule). For any integer n , if $f(x) = x^n$ then $f'(x) = n x^{(n-1)}$.

Proof. Expand $(x+h)^n$ by the binomial theorem and cancel the leading term, then take the limit as h approaches 0.

Example 3.6. Differentiate $g(x) = x^3$. Applying the power rule gives $g'(x) = 3 x^2$.

Exercises

1-3. Differentiate each function.

1. $f(x) = x^5$

2. $f(x) = \sin x$

3. $f(x) = e^x$